



Coordinate Geometry

✧✧ The Ultimate Studygram Cheat Sheet ✧✧



Distance

Section

Mid-point



The Grid Basics



X-Axis = Abscissa
(horizontal distance)



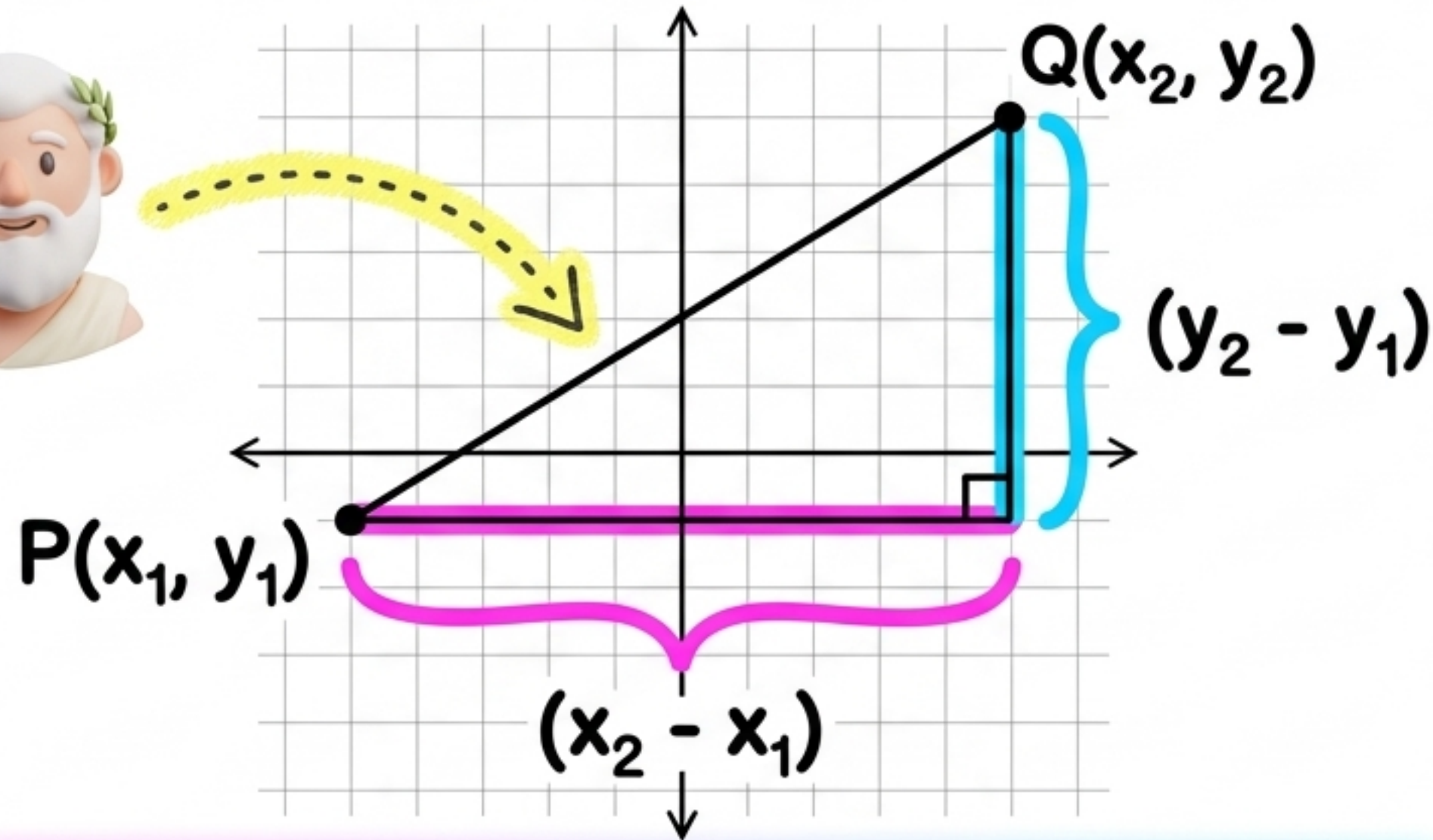
Y-Axis = Ordinate
(vertical distance)



Coordinates always live in pairs: (x, y)! 

Points on x-axis: (x,0) | Points on y-axis: (0,y)

Distance Formula



$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

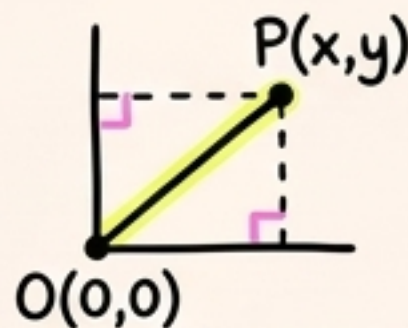
The Origin Shortcut



Distance from the
Origin (0,0)?



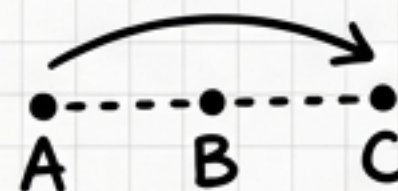
$$\sqrt{(x^2 + y^2)}$$



Exam Cheat Sheet: Proving Shapes

- **Collinear:**

$$AB + BC = AC$$



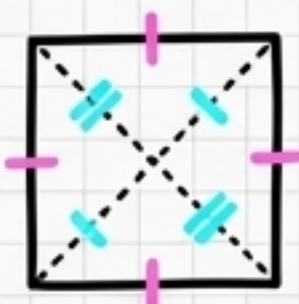
- **Isosceles:**

2 sides equal

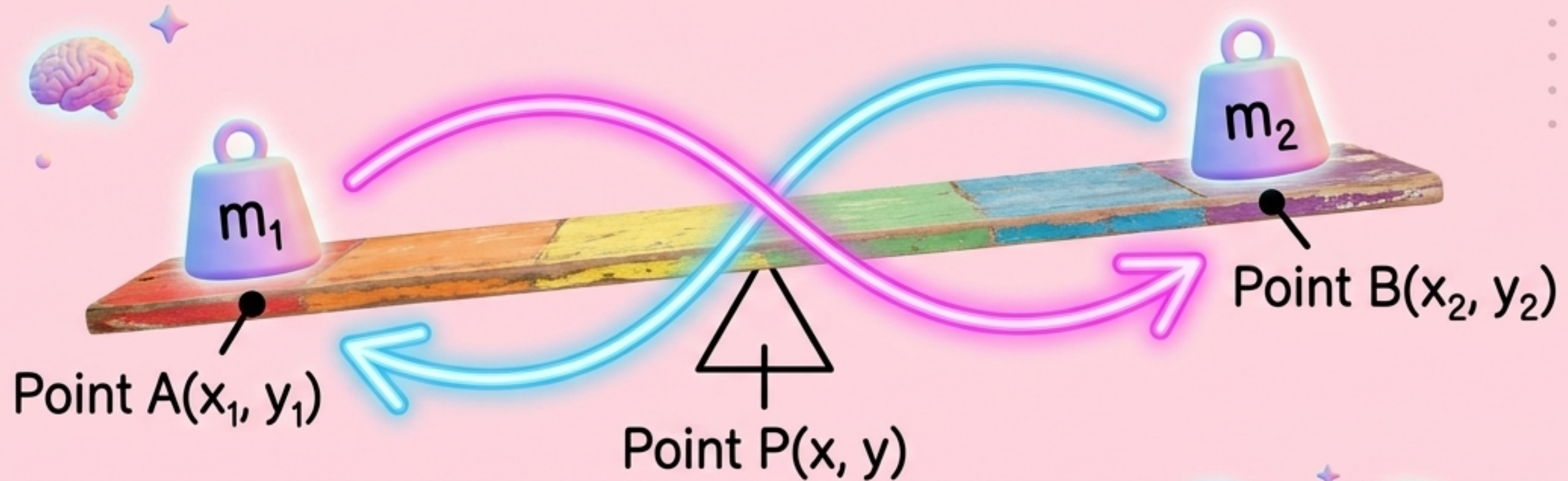


- **Square:**

4 sides equal AND
diagonals equal



Section Formula



$$x = (m_1x_2 + m_2x_1) / (m_1 + m_2)$$
$$y = (m_1y_2 + m_2y_1) / (m_1 + m_2)$$

Mid-Point Magic ✨

What if $m_1 = m_2$? The ratio is 1:1!



$$\text{Mid-point } P = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Hack: Just
average the x's,
average the y's!
Easy peasy! 😊

Exam Hack: Finding the Ratio

Don't use $m_1:m_2$!



Let the ratio be $k:1$!



Use the section formula
and solve for k .



Pro Tip! If point P is on the x-axis, its y-coordinate is 0.
Set the y-formula to 0 to instantly solve for k !

Chapter Summary 📷 (Save this!)

Mid-Point

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Distance

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Origin} = \sqrt{x^2 + y^2}$$

Section

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$